

Revised Load Interconnection System Impact Study Report

For Queue Project Q18-1071

Skaggs Island Junction 2 345 kV
September 22, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1071. The Interconnection Customer proposes a 633.5 MW large load (data center) facility interconnecting at the Skaggs Island Junction 2 345 kV substation in the Virginia area. The request was received on October 03, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 437 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$78,666,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CHIPOLA to WESTCOAST CELLUFIBRE 138 kV	LEAVITT to CHANCELOR ROW 345 kV	126.3	358	6.8
IMPERIAL to EUFAULA 500 kV	TREGO to GALION GENERATING STTION 500 kV	119.7	4,273	8.4
EUFAULA to BEAVER 500 kV	GALION GENERATING STA to TREGO 500 kV	115.6	5,086	8.7
SKAGGS ISLAND JCT 2 to CHANCELLOR ROW 345 kV	GASTON to TECUMSEH ENERGY CENTER 345 kV	115.3	1,175	64.9
TREGO to GALION GENERATING STA 500 kV	BEAVER to TREGO 500 kV	113.4	3,288	5.3
TREGO to BEAVER 500 kV	ADAMS to ONANCOCK DIESEL GENERATION FACILITY SUBSTATION 230 kV L	113.1	5,833	8.9
TREGO to TROY ENERGY LLC SUBSTATION 500 kV L	TREGO to SWAN VALLEY 500 kV	112.1	4,121	8.3
ARTESIA COUNTRY CLUB SUBSTATION to AIR FORCE BASE 345 kV	JACK MCDONOUGH to LOW GAP SUBSTATION 138 kV	111.3	1,457	10.2
SWAN VLY to MUSKOGEE 500 kV	RICHLAND to DEKALB 500 kV	111.3	4,382	5.3
TIDD SUBSTATION to MUSKINGM RIV 500 kV line	EL PASO to RAVENNA SUBSTATION 500 kV	109.8	2,844	6.5
RICHLAND SUBSTATION to DEKALB 500 kV	SWAN VLY to TREGO 500 kV	109.6	3,847	9.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FORT GIBSON to ARMSTRONG 500 kV	HILLTOP to PINEY CRK PROJECT SUBSTATION 500 kV	109.0	2,513	6.0
WESTCOAST CELLUFIBRE to LEAVITT 138 kV Ckt 1	CHANCELLOR ROW SUBSTATION to SKAGGS ISLAND JCT 2 345 kV	107.6	281	9.2
WESTCOAST CELLUFIBRE to LEAVITT 138 kV Circuit 2	SKAGGS ISLAND JUNCTION 2 to CHANCELLOR ROW 345 kV	107.6	281	9.2
BURWELL to P.T.S 345 kV	ALLEGHENY ENERGY UNITS 3 4 & 5 345/1 kV Transformer	107.2	1,721	5.3
FT HUMBUG to ORCHARD ROAD 500 kV Line	WEST CAMPUS to FT HUMBUG 345 kV Line	106.2	4,302	6.8
DRESDEN SUBSTATION to CONESVILLE STA 500 kV	TREGO to BEAVER 500 kV L	105.4	3,582	5.2
SUGAR HILL to ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV	PINEY CRK PROJECT to HILLTOP 500 kV	105.0	2,356	6.8
ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV/1 kV Auto	HILLTOP to PINEY CREEK PROJECT 500 kV	105.0	2,177	6.6
ALLEGHENY ENERGY UNITS 3 4 & 5 500 kV/1 kV TX	PINEY CREEK PROJECT to HILLTOP 500 kV	105.0	2,177	6.6
SWAN VLY to TREGO 500 kV	RICHLAND to TROY ENERGY LLC 500 kV	104.7	5,841	5.1
RICHLAND SUBSTATION 500-230 kV XFMR	TROY ENERGY LLC to RICHLAND 500 kV	104.5	2,150	6.3
CLARK COUNTY to ARTESIA COUNTRY CLUB 345 kV L	JACK MCDONOUGH SUBSTATION to LOW GAP 138 kV line	104.1	1,761	8.0
TECUMSEH ENERGY CTR to GASTON SUBSTATION 345 kV	CLOVER to BEAR 345 kV	103.4	1,905	6.2
CALDWELL to PINEY CREEK PROJECT 345 kV	PINEY CRK PROJECT to HILLTOP 500 kV Line	103.0	2,712	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WESTCOASTCELLUFIBRE to CHIPOLA 138 kV line	Rebuild line with higher-capacity conductor	\$2,359,000	21% (\$498,000)
IMPERIAL to EUFAULA 500 kV	Rebuild line with higher-capacity conductor	\$82,389,000	3% (\$2,770,000)
BEAVER to EUFAULA 500 kV	Rebuild line with higher-capacity conductor	\$73,677,000	3% (\$2,476,000)
SKAGGS ISLAND JUNCTION 2 to CHANCELLOR ROW 345 kV	Rebuild line with higher-capacity conductor	\$17,419,000	43% (\$7,517,000)
TREGO to GALION GENERATING STA 500 kV	Rebuild line with higher-capacity conductor	\$193,747,000	3% (\$6,555,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BEAVER to TREGO 500 kV	Rebuild line with higher-capacity conductor	\$776,751,000	3% (\$26,072,000)
TREGO to TROY ENERGY LLC SUBSTATION 500 kV L	Rebuild line with higher-capacity conductor	\$145,834,000	3% (\$4,915,000)
ARTESIA COUNTRY CLUB SUBSTATION to AIR FORCE BASE 345 kV	Rebuild line with higher-capacity conductor	\$30,382,000	5% (\$1,489,000)
SWAN VLY to MUSKOGEE 500 kV	Rebuild line with higher-capacity conductor	\$178,180,000	5% (\$8,360,000)
MUSKINGUM RIV SUBSTATION to TIDD 500 kV	Reconductor line	\$121,207,000	3% (\$4,023,000)
RICHLAND SUBSTATION to DEKALB 500 kV	Reconductor line	\$61,577,000	3% (\$2,113,000)
FORT GIBSON to ARMSTRONG 500 kV	Reconductor line	\$35,435,000	3% (\$1,213,000)
WESTCOAST CELLUFIBRE to LEAVITT 138 kV Ckt 1	Reconductor line	\$537,000	28% (\$148,000)
WESTCOAST CELLUFIBRE to LEAVITT 138 kV Circuit 2	Reconductor line	\$530,000	28% (\$146,000)
BURWELL to P.T.S 345 kV	Reconductor line	\$3,581,000	4% (\$146,000)
ORCHARD RD to FORT HUMBUG 500 kV	Reconductor line	\$23,515,000	3% (\$680,000)
DRESDEN to CONESVILLE STA 500 kV	Reconductor line	\$12,623,000	5% (\$578,000)
SUGAR HILL to ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV	Reconductor line	\$6,698,000	3% (\$222,000)
ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV/1 kV Auto	Replace transformer with higher-rated unit	\$17,998,000	3% (\$596,000)
ALLEGHENY ENERGY UNITS 3 4 & 5 500 kV/1 kV TX	Replace transformer with higher-rated unit	\$31,729,000	3% (\$1,050,000)
SWAN VLY to TREGO 500 kV	Reconductor line	\$35,632,000	6% (\$2,076,000)
RICHLAND SUBSTATION 500-230 kV XFMR	Replace transformer with higher-rated unit	\$23,107,000	3% (\$793,000)
ARTESIA COUNTRY CLUB to CLARK COUNTY 345 kV	Reconductor line	\$15,961,000	5% (\$830,000)
TECUMSEH ENERGY CTR to GASTON SUBSTATION 345 kV	Reconductor line	\$34,994,000	5% (\$1,673,000)
PINEY CREEK PROJECT to CALDWEL 345 kV	Reconductor line	\$39,695,000	4% (\$1,727,000)
	Total allocated to project		\$78,666,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.